

Matthew Hurtado

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Research-oriented Computer Science and Mathematics dual major with hands-on experience in high-performance computing and machine learning. Proven ability to design and implement complex simulations using ROSS and MPI, and to build and train transformer models from the ground up. Seeking to apply a strong foundation in numerical optimization and parallel computing to a challenging research or engineering role.

Education

Rensselaer Polytechnic Institute (RPI)	<i>Expected Graduation: December 2025</i>
Dual Major - Bachelor of Science in Computer Science & Mathematics, Minor in Logic	<i>Troy, NY</i>
Cumulative GPA: 3.76	
Rensselaer Leadership Award, Chess Club, BJJ Club	
New York Kappa, Pi Mu Epsilon	<i>December 2024 - Present</i>
Member	<i>Troy, NY</i>

Experience

Rensselaer Polytechnic Institute	<i>Aug 2023 - Present</i>
Teaching Assistant & Tutor	<i>Troy, NY</i>
- Lead weekly recitation sessions for ~30 students and provide one-on-one support across foundational and advanced courses, including Computer Science I, Foundations of Computer Science, and Mathematical Foundations of Machine Learning. - Tutor students in multivariable calculus, differential equations, algorithms, and numerical computing, reinforcing rigorous mathematical reasoning and problem-solving strategies. - Support student development in proof-writing, algorithm design, and technical intuition through personalized academic assistance and structured walkthroughs.	
Rho Pi, Rensselaer Alpha Epsilon Pi Fraternity	<i>Jan 2023 - Present</i>
Member	<i>Troy, NY</i>

Relevant Projects

Parallel Discrete-Event Simulation of Pandemic Dynamics using ROSS and MPI	<i>Spring 2025</i>
- Authored performance methodology and final report for a large-scale agent-based pandemic simulation. - Analyzed simulation performance on the AiMOS supercomputer, scaling the model to 524,288 agents and evaluating strong/weak scaling characteristics using ROSS and MPI.	
ByteLM — A Byte-Level Language Model	<i>Spring 2025</i>
- Built ByteLM, a 58.6M-parameter transformer trained from scratch with byte-level tokenization on a single RTX 4060 Ti over 72 hours. - Integrated standard attention and experimented with scaling, curriculum learning, and dataset cleaning. - Published source code and training logs; project focused on architecture simplicity and self-hosted training.	
Financial Market Prediction with Autoregressive Models	<i>Summer - Fall 2024</i>
- Implemented transformer-based models to predict financial market closing prices. - Compared autoregressive models, including linear regression, ARMA, RNNs, and transformers, for time-series prediction. - Developed educational modules and published code on GitHub to teach autoregressive modeling techniques.	

Skills

- Programming Languages:** Python, C++, C, Java, C#, SQL, MATLAB, MIPS Assembly
- Machine Learning & Data Science:** PyTorch, NumPy, scikit-learn, Pandas, TA-Lib, Qiskit
- High-Performance Computing:** MPI, CUDA, PDES (ROSS)
- Mathematical & Computational Concepts:** Time Series Forecasting, Numerical Optimization, Parallel Computing, Data Structures & Algorithms
- Productivity Tools:** Git, Excel, Bash, L^AT_EX